

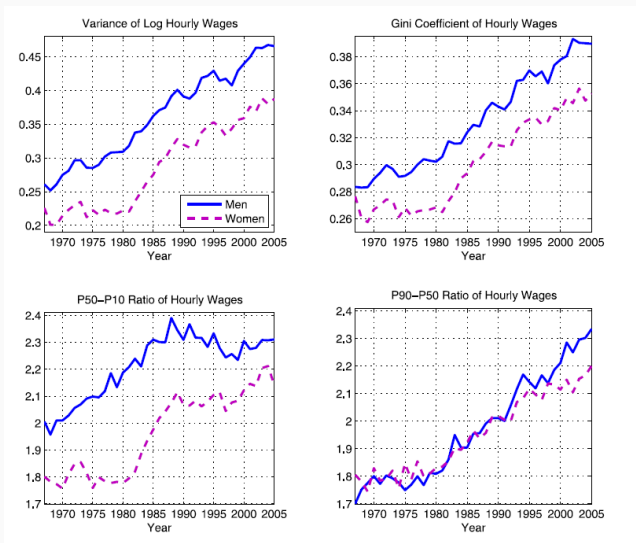
Macroeconomics II, Lecture XI:

Incomplete markets models: basics

Erik Öberg

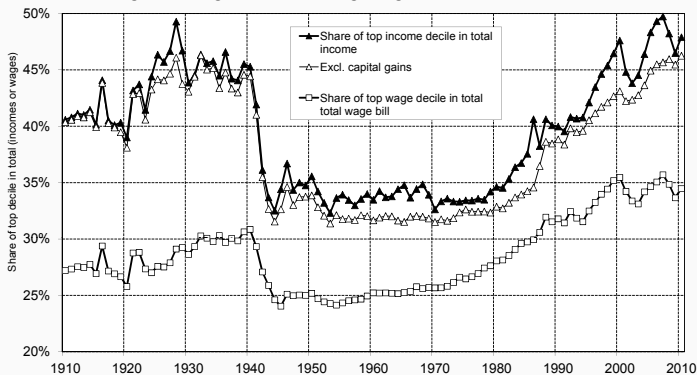
- Traditional macroeconomic models assume a representative agent: **no room for studying the interaction between macroeconomic dynamics and inequality**
- Inequality and economic heterogeneity are, however, very salient features of the macroeconomic environments we seek to understand
- In the last part of this course, we will study macroeconomic models with **heterogeneous households**
 - Alternatively, macroeconomic models with **incomplete asset markets**
- We use these models to investigate:
 1. the distributional implications of macroeconomic events
 - Does technical change increase income inequality?
 - Does wealth inequality widen when credit becomes cheaper?
 2. how household heterogeneity matters for macroeconomic transmission
 - Does wider wealth inequality imply more business cycle volatility?
 - Does the existence of more high-debt households imply that monetary policy is more potent?

US cross-sectional wage inequality



From Heathcote-Perri-Violante (RED 2010) using CPS data

Figure 8.7. High incomes and high wages in the U.S. 1910-2010

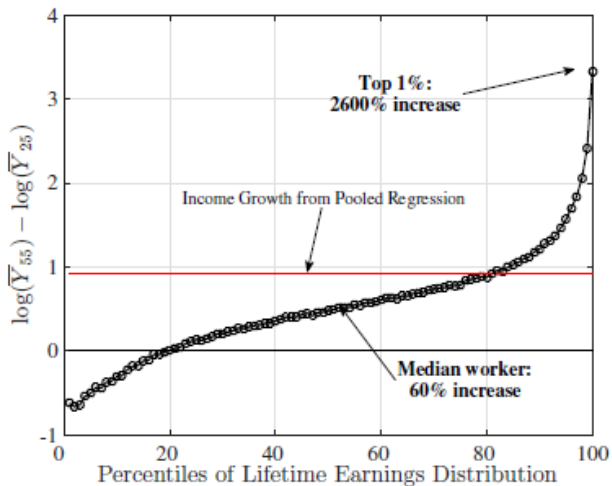


The rise of income inequality since the 1970s is largely due to the rise of wage inequality.

Sources and series: see piketty.pse.ens.fr/capital21c.

From Piketty (Book 2014) using US register data

US income inequality over the life-cycle



From Guvenen-Karahan-Ozkan-Song (Ecmtra 2021) using US register data

Income inequality widens in recessions

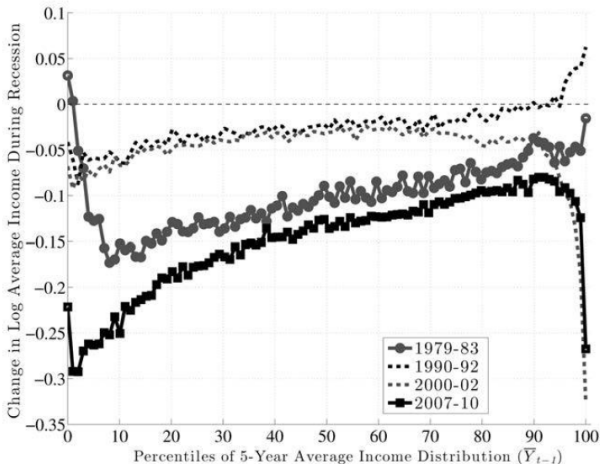
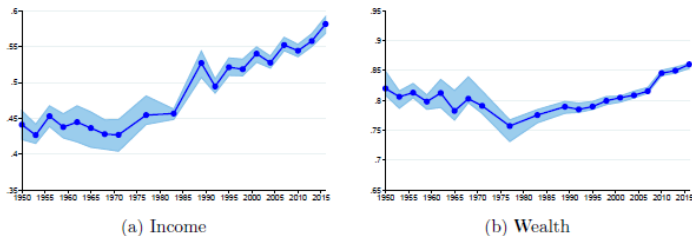


FIG. 13.—Change in log average earnings during recessions, prime-age males

From Guvenen-Ozkan-Song (JPE 2014) using US register data

Figure 4: Gini coefficients for income and wealth with confidence bands



From Kuhn-Schularick-Steins (JPE 2020) using SCF+ data

- Lecture 11: Incomplete markets models: basics
- Lecture 12: Buffer-stock savings: model meets data
- Lecture 13: Incomplete markets in general equilibrium

1. Aggregation under different market structures
 - Autarky
 - Complete markets
 - Incomplete markets
2. Consumption-savings dynamics with incomplete markets
 - The household problem: setup and characterization
 - Precautionary savings

Aggregation and market structure

- The interaction between aggregate and distributional dynamics depends on the underlying **market structure**
- **Market structure** refers to which assets are available to the household; it determines how the household can allocate resources across states.
- The household budget constraint in three benchmark structures:
 1. **Autarky:**

$$C_{it}(s^t) \leq Y_{it}(s^t) \quad \forall t, s^t$$

2. **Complete markets:**

$$C_{it}(s^t) + \sum_{s_{t+1}|s^t} Q_t(s^t, s_{t+1}) A_{it+1}(s^t, s_{t+1}) \leq Y_{it}(s^t) + A_{it}(s^t, s_t) \quad \forall t, s^t$$

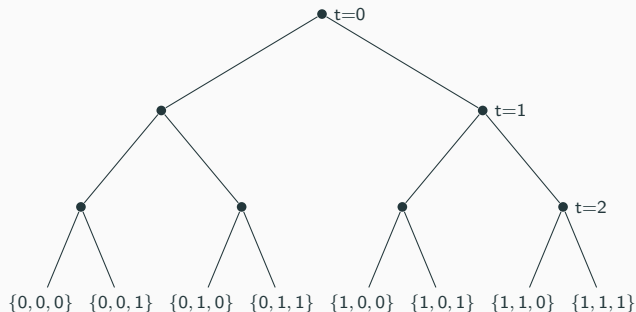
3. **(Standard) Incomplete markets:**

$$C_{it}(s^t) + Q_t(s^t) A_{it+1}(s^t) \leq y_{it}(s^t) + A_{it}(s^{t-1}) \quad \forall t, s^t$$

- Time is discrete and indexed by t
- The economy has N households indexed by i
- All households live for T periods; $T = \infty$ is possible
- One non-storable consumption good
- In each period, an aggregate event $s_t \in S$ is realized
- At time $t = 0$, each particular event history at time t , $s^t = \{s_0, s_1, \dots, s_t\} \in S^t$, is realized with probability $\pi(s^t)$
- Given event history s^t at time t , each event at time $t + 1$, $s_{t+1} \in S$, is realized with probability $\pi(s_{t+1}|s^t)$
- For each event history s^t , household i receives endowment $Y_t^i(s^t)$

Example event history

- Assume $s_t \in S = \{0, 1\}$
- In period 3, there are 8 possible “event histories” or “states of the world”:



- Note 1: Here, individual endowment depends on the entire event history,
 $Y_t^i = Y_t^i(s^t)$
- Note 2: Even if income only depends on the realized shock today,
 $Y_t^i = Y_t^i(s_t)$, consumption and savings choices in period t depend on the entire event history s^t

- All households have identical preferences: time-separable, common discount factor β
- Rational expectations: subjective beliefs about event probabilities coincide with actual probabilities
- Household objective

$$\begin{aligned}U_i &= E_0 \sum_{t=0}^T \beta^t U(C_{it}(s^t)) \\&= \sum_{t=0}^T \sum_{s^t \in S^t} \beta^t \pi(s^t) U(C_{it}(s^t))\end{aligned}$$

- U has the usual regularity conditions: twice differentiable, strictly increasing, strictly concave and satisfies the Inada conditions

- A consumption allocation is a collection $\{C_{it}(s^t)\}_{i=1}^N$
- A consumption allocation is feasible if

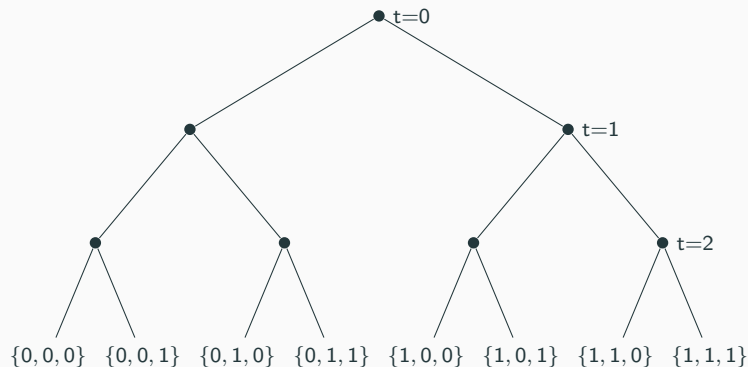
$$\begin{aligned}C_{it}(s^t) &\geq 0 \quad \forall i, t, s^t \\C_t(s^t) &\leq Y_t(s^t) \quad \forall t, s^t\end{aligned}$$

where

$$C_t(s^t) \equiv \sum_{i=1}^N C_{it}(s^t), \quad Y_t(s^t) \equiv \sum_{i=1}^N Y_{it}(s^t)$$

- A feasible consumption allocation $\{C_{it}(s^t)\}_{i=1}^N$ is Pareto efficient if there is no other feasible consumption allocation $\{\hat{C}_{it}(s^t)\}_{i=1}^N$ such that

$$\begin{aligned}U(\hat{C}_{it}(s^t)) &\geq U(C_{it}(s^t)) \quad \text{for all } i \\U(\hat{C}_{it}(s^t)) &> U(C_{it}(s^t)) \quad \text{for some } i\end{aligned}$$



- Autarky: nothing can be traded at any node

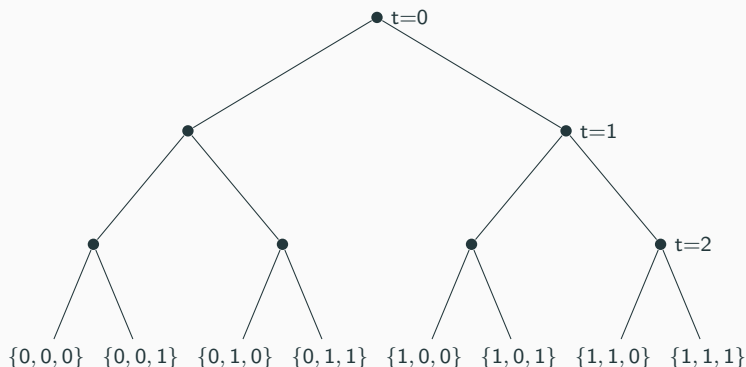
- Household problem:

$$\begin{aligned} \max_{c(s^t)} \quad & \sum_{t=0}^T \sum_{s^t \in S^t} \beta^t \pi(s^t) U(C_{it}(s^t)) \\ \text{s.t.} \quad & c_{it}(s^t) \leq y_{it}(s^t) \quad \forall t, s^t \end{aligned}$$

- Solution to household problem is trivial: $c_{it}(s^t) = y_{it}(s^t)$
- Distributional dynamics follows mechanically: individual consumption tracks individual income
- Aggregate dynamics follows mechanically: no prices, and

$$C_t(s^t) = Y_t(s^t) \quad \forall t, s^t$$

Complete markets (with Arrow securities)



- At each node s^t , two assets can be traded:
 - pays off one consumption good at node $\{s^t, 0\}$ and costs $Q_t(s^t, 0)$
 - pays off one consumption good at node $\{s^t, 1\}$ and costs $Q_t(s^t, 1)$

- Household problem:

$$\begin{aligned} \max_{C_{it}(s^t), A_{it+1}(s^t, s_{t+1})} \quad & \sum_{t=0}^T \sum_{s^t \in S^t} \beta^t \pi(s^t) U(C_{it}(s^t)) \\ \text{s.t.} \quad & C_{it}(s^t) + \sum_{s_{t+1} | s^t} Q_t(s^t, s_{t+1}) A_{it+1}(s^t, s_{t+1}) \leq \\ & Y_{it}(s^t) + A_{it}(s^t, s_t) \quad \forall t, s^t \\ & A_{it+1}(s^t, s_{t+1}) \geq -\bar{A}_i \quad \forall t, s^t \end{aligned}$$

- The **First welfare theorem** applies: the allocation is Pareto efficient and therefore coincides with that of a social planner for some Pareto weights $\{\alpha_i\}_1^N$

- Equilibrium features **full insurance**: (Do on whiteboard)

$$\frac{U_c(C_{it}(s^t))}{U_c(C_{jt}(s^t))} = \frac{\alpha_j}{\alpha_i} \quad \forall s^t$$

- Full insurance: if my gain from consuming more in some state is high, yours must be high in that state too
- With CRRA preferences $U(C_{it}(s^t)) = \frac{(C_{it}(s^t))^{1-\sigma}-1}{1-\sigma}$, full insurance implies that the allocation and prices only depend on aggregate consumption:

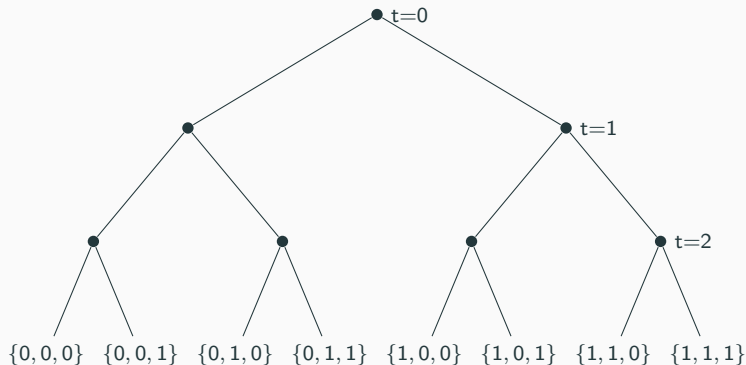
$$\begin{aligned} C_{it}(s^t) &= \theta_i C_t(s^t) \\ Q_t(s^t, s_{t+1}) &= \beta \pi(s_{t+1}|s^t) \frac{U_c(C_{t+1}(s^{t+1}))}{U_c(C_t(s^t))} \end{aligned}$$

- Hence, the distribution of consumption is constant over time...
- ... and aggregate dynamics are **as if** there were a representative agent who solves

$$\begin{aligned} \max_{C_t(s^t), A_{t+1}(s^t)} \quad & \sum_{t=0}^T \sum_{s^t \in S^t} \beta^t \pi(s^t) U(C_t(s^t)) \\ \text{s.t.} \quad & C_t(s^t) + \sum_{s_{t+1} | s^t} Q_t(s^t, s_{t+1}) A_{t+1}(s^t, s_{t+1}) \\ & \leq Y_t(s^t) + A_t(s^t, s_{t-1}) \quad \forall t, s^t \\ & A_{t+1}(s^t) \geq -\bar{A} \quad \forall t, s^t \end{aligned}$$

- $\Rightarrow$ If markets are sufficiently complete, rep-agent macro seems legitimate
 - A large literature tests the extent of consumption insurance; see Violante's textbook chapter

(Standard) Incomplete markets



- At each node s^t , one asset that pays off one consumption good in period $t + 1$ and costs $Q(s^t)$ can be traded

- Household problem:

$$\begin{aligned} \max_{C_{it}(s^t), A_{it+1}(s^t)} \quad & \sum_{t=0}^T \sum_{s^t \in S^t} \beta^t \pi(s^t) U(C_{it}(s^t)) \\ \text{s.t.} \quad & C_{it}(s^t) + Q_t(s^t) A_{it+1}(s^t) \leq Y_{it}(s^t) + A_{it}(s^{t-1}) \quad \forall t, s^t \\ & A_{it+1}(s^t) \geq -\bar{A}_i \quad \forall t, s^t \end{aligned}$$

- Due to lack of insurance, ex-ante homogeneous households will be **ex-post heterogeneous** in terms of $\{C, A, Y\}$
- Due to lack of insurance, the allocation is generally inefficient
- The remainder of this course is devoted to explaining these two claims and deriving their implications

Consumption-savings dynamics with incomplete markets

The Income-Fluctuations Problem

- We will organize our study around the [income-fluctuations problem](#)
- Same household problem as above:

$$\begin{aligned} \max_{C_{it}(s^t), A_{it+1}(s^t)} \quad & \sum_{t=0}^T \sum_{s^t \in S^t} \beta^t \pi(s^t) U(C_{it}(s^t)) \\ \text{s.t.} \quad & C_{it}(s^t) + Q_t(s^t) A_{it+1}(s^t) \leq Y_{it}(s^t) + A_{it}(s^{t-1}) \quad \forall t, s^t \\ & A_{it+1}(s^t) \geq -\bar{A}_i \quad \forall t, s^t \end{aligned}$$

For simplicity, we

- do not keep track of the full history s^t
- assume assets cost 1 and pay R consumption goods (instead of costing Q_t and paying 1 consumption good)
- drop household index i , as we will only study one household
- assume a tight credit constraint $\bar{A} = 0$
- assume income is i.i.d.: $Y_t = \bar{Y} + \epsilon_t$ with $\epsilon_t \sim F$, where F has finite support $[\epsilon_{min}, \epsilon_{max}]$

- Household problem

$$\begin{aligned} \max_{C_t, A_{t+1}} \quad & E_0 \sum_{t=0}^{\infty} \beta^t U(C_t) \\ \text{s.t.} \quad & C_t + A_{t+1} \leq Y_t + RA_t \\ & A_{t+1} \geq 0 \end{aligned}$$

- For some applications, we will work with a **recursive** formulation of this problem
- Recursive household problem: find value function V and policy functions C, A' that solve

$$\begin{aligned} V(A, Y) &= \max_{C, A'} U(C) + \beta EV(A', Y') \\ \text{s.t.} \quad & C + A' \leq Y + RA \\ & A' \geq 0 \end{aligned}$$

- We have written this problem as if there are two **state variables**
- Actually, for i.i.d. shocks, there is only one

- Our problem

$$\begin{aligned} V(A, Y) &= \max_{C, A'} U(C) + \beta EV(A', Y') \\ \text{s.t.} \quad & C + A' \leq Y + RA \\ & A' \geq 0 \end{aligned}$$

- When studying a recursive problem, always ask yourself what the minimal set of state variables is!
 - Here, we seemingly have two state variables: $\{A, Y\}$
 - But, to choose how much to consume/save, the household only needs to know its total **cash on hand** $M = Y + RA$
 - And conditional on knowing M , knowing A or Y does not help to forecast M' and therefore neither the continuation value $EV(\cdot)$
 - Note: this is only true when Y_t is i.i.d.
 - Not true with persistent shocks! E.g. if $Y' = \rho Y + \epsilon$

- Reformulating our constraints with cash-on-hand $M = Y + RA$ and without A

$$C + A' \leq Y + RA$$

$$\Rightarrow A' \leq M - C$$

$$\Rightarrow M' \leq R(M - C) + Y'$$

and

$$A' \geq 0$$

$$\Rightarrow C \leq M$$

- Reformulated recursive problem:

$$\begin{aligned} V(M) &= \max_{C, M'} U(C) + \beta EV(M') \\ \text{s.t.} \quad &M' \leq R(M - C) + Y' \\ &C \leq M \end{aligned}$$

- Solution given by value function $V(M)$ and consumption policy function $C(M)$

1. Solution is characterized by an Euler equation,

$$U_c(C) = \beta RE [V_m(M')] + \mu$$

2. complementary slackness conditions,

$$\mu(C - M) = 0$$

$$\mu \geq 0$$

$$\lambda(M' - R(M - C) - Y') = 0$$

$$\lambda \geq 0$$

Note:

- Non-satiated preferences imply that budget constraint always binds: $\lambda > 0$.
- The credit constraint may either
 - 2.1 bind $\Rightarrow$ corner solution with $\mu > 0$
 - 2.2 not bind $\Rightarrow$ interior solution with $\mu = 0$

3. and an envelope condition (Do on whiteboard)

$$V_m(M') = U_c(C').$$

- Put together, an interior solution satisfies

$$\begin{aligned}U_c(C(M)) &= \beta REV_m(R(M - C(M) + Y')) \\ &= \beta REU_c(C')\end{aligned}$$

- a corner solution satisfies

$$C(M) = M$$

- If interior solution is feasible for M , then also feasible for any $\hat{M} > M$
- If interior solution not feasible for M , then neither feasible for any $\hat{M} < M$
- Therefore, the global solution $C = C(M)$ satisfies

$$C(M) = \begin{cases} M & \text{if } M \leq M^* \\ (U_c)^{-1} [\beta REU_c(C')] & \text{if } M > M^* \end{cases}$$

for some cutoff M^*

- So far: how to characterize consumption-savings problems with uninsurable income risk
- Now: what does uninsurable income risk imply for consumption-savings dynamics?
- One lesson we've already learned: households may face a binding credit constraint, in which case they have a high **marginal propensity to consume (MPC)**
- Now: uninsurable income risk produces an additional savings motive: the **precautionary-savings motive**

- With uninsurable income risk, **precautionary savings** arise if
 1. Household preferences exhibit **prudence**, or
 2. Households face a potentially binding **credit constraint**

- Prudent preferences: preferences with $U_{ccc} > 0$
- Consider again our household problem:

$$\begin{aligned} V(M) &= \max_{C, M'} U(C) + \beta EV(M') \\ \text{s.t.} \quad & M' \leq R(M - C) + Y' \\ & C \leq M \end{aligned}$$

- Without loss of generality, assume $\beta R = 1$
- Assume the credit constraint is never binding; the problem is characterized by

$$U_c(C) = EU_c(C')$$

- Suppose $U_{ccc} = 0$ (as with quadratic preferences). Then U_c is a linear function, and

$$\begin{aligned}U_c(C) &= E_t U_c(C') \\ &= U_c(EC')\end{aligned}$$

implying $C = EC'$

- Now suppose $U_{ccc} > 0$ (as with CRRA preferences). Then U_c is convex
- Using Jensen's inequality, we have

$$\begin{aligned}U_c(C) &= EU_c(C') \\ &> U_c(EC')\end{aligned}$$

implying $C < EC'$

- Lesson: for the same income stream, prudent preferences imply that consumption is expected to grow over time $\Rightarrow$ the household saves more today!

- Preferences that exhibit prudence:
 - CRRA: $U(C) = \frac{C^{1-\sigma}}{1-\sigma}$
 - CARA: $U(C) = 1 - \frac{1}{\gamma} e^{-\gamma C}$
 - Most other preferences you see in macro models
- Preferences that do not exhibit prudence:
 - Linear: $U(C) = C$
 - Quadratic: $U(C) = -C^2 + \bar{C}$
- Standard reference: Carroll-Kimball (Ecmtra 1996)

- Precautionary savings may also arise due to a **potentially binding credit constraint**
- Our household problem

$$\begin{aligned} V(M) &= \max_{C, M'} U(C) + \beta EV(M') \\ \text{s.t.} \quad &M' \leq R(M - C) + Y' \\ &C \leq M \end{aligned}$$

- As before, assume $\beta R = 1$ without loss of generality
- Euler equation and complementary slackness:

$$\begin{aligned} U_c(C) &= \beta R E U_c(C') + \mu \\ \mu(C - M) &= 0 \\ \mu &\geq 0 \end{aligned}$$

- For this exercise, I reintroduce time subscripts
- Assume quadratic preferences, such that prudence-induced savings are excluded. Euler equation:

$$-C_t = -EC_{t+1} + \mu_t$$

or

$$C_t = EC_{t+1} - \mu_t$$

- If the credit constraint is not binding in period t : $C_t = EC_{t+1}$
- If binding: $C_t < EC_{t+1}$
- If the credit constraint is binding, the household consumes less today than it would like given its consumption-smoothing motive

- Suppose the credit constraint is not binding today, but that it might be binding tomorrow:

$$\begin{aligned}C_t &= E_t C_{t+1} \\ C_{t+1} &= E_{t+1} C_{t+2} - \mu_{t+1}\end{aligned}$$

- By the law of iterated expectations

$$C_t = E_t [C_{t+2}] - E_t [\mu_{t+1}]$$

- If $\mu_{t+1} > 0$ in some states of the world, then $E_t [\mu_{t+1}] > 0$
- Compare to the solution when the constraint is never binding:

$$C_t = E_t [C_{t+2}]$$

- Lesson: anticipated future binding constraints reduce consumption today relative to the unconstrained consumption-smoothing benchmark
 - I.e., anticipating a binding credit constraint induces the household to save more
 - By saving more, the household can bring consumption closer to its unconstrained optimum in the states where the constraint binds

- Previous example had constraint $A_{t+1} \geq 0$
- More general credit constraint $A_{t+1} \geq -\bar{A}$
- Which values of $\bar{A}$ are allowed?
- See your problem set

- The precautionary-savings motive is intimately linked to the [marginal propensity to consume \(MPC\)](#)
 - We'll talk about this more in the next lecture
- Prudence and potentially binding credit constraints can interact in non-trivial fashion; see Carroll-Kimball-Holm (JET 2021)
- Another standard reason why income risk may affect consumption dynamics: irreversible durable purchases
 - See Bernanke (AER 1982); Dixit-Pindyck (Book 1994); Harmenberg-Öberg (JME 2021)

- Complete markets: ex-ante homogeneous households will be ex-post homogeneous
- Incomplete markets: ex-ante homogeneous households will be ex-post heterogeneous
- Much of the incomplete-markets literature is organized around the income-fluctuations problem
- Features of the income-fluctuations problem:
 - Stochastic income
 - Limited set of assets (typically: one risk-free asset)
 - Potentially binding credit constraint
- With incomplete markets, households have a precautionary-savings motive
 - Prudence
 - Potentially binding credit constraint